VEHICLE-GRID INTEGRATION COUNCIL

May 15th, 2026

VIA BPU E-FILING SYSTEM AND ELECTRONIC MAIL

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RE: Vehicle Grid Integration Council Comments in the Matter of Developing Integrated Distributed Energy Resource Plans to Modernize New Jersey's Electric Grid, Docket No. QO24030199

Vehicle-Grid Integration Council ("VGIC")[1] is a 501(c)(6) membership-based advocacy organization committed to advancing the role of vehicle-grid integration ("VGI") through policy development, education, and research. VGIC supports the transition to decarbonized transportation and electric sectors by ensuring that the value from flexible electric vehicle ("EV") charging and discharging is recognized, compensated, and aligned with a more reliable, affordable, and efficient electric grid.

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[1] VGIC member companies and supporters include American Honda Motor Co., Inc., Ford Motor Company, General Motors, Nissan Group of North America, Bidirectional Energy, BMW of North America LLC, bodeEV, ChargePoint, ChargeScape, Codibly Inc., Critical Loop, dcbel, Eaton Corporation, EnergyHub, EV.Energy, Fermata Energy, Gravity Inc., Mercedes-Benz Research & Development North America, Inc., Orange Charger, Inc., Renew Home, Rivian, Stellantis, SWTCH Energy Inc., Tesla, The Mobility House, Toyota Motor North America, Inc., UL Solutions, WeaveGrid, Clean Power Alliance, Connecticut Green Bank, Los Angeles Department of Water and Power, Marin Clean Energy, San Diego Community Power, and Sacramento Municipal Utility District. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all individual VGIC member companies or supporters. (https://www.vgicouncil.org/)

Introduction

VGIC commends the New Jersey Board of Public Utilities ("BPU") for initiating this important proceeding to advance integrated distributed energy resource planning, modernize interconnection processes, and identify opportunities to more efficiently and cost-effectively enable the interconnection of distributed energy resources ("DERs") across the State. In particular, VGIC recognizes the Board's focus on improving hosting capacity transparency, addressing constrained circuits, and exploring regulatory and technical pathways to accelerate DER deployment in response to growing load and affordability pressures.

VGIC appreciates the opportunity to provide comments and share its informed perspective based on extensive work across the United States on vehicle-grid integration potential and bidirectional charging system interconnection. VGIC member companies are actively advancing these solutions and bring direct experience working with utilities, regulators, and system operators to enable the safe, reliable, and scalable integration of electric vehicles as grid resources.

The focus of our comments is on EVs as flexible grid resources and the opportunity to leverage the latent energy storage capacity in EVs to address the energy affordability challenge. Research from New York[2] and Massachusetts[3], among others, finds that electric vehicles represent the single largest source of flexible demand on the system, driven by their scale, high availability, and multi-kilowatt capability. Compared to traditional curtailment-based demand response resources, these characteristics allow EVs to deliver capacity more efficiently and with lower administrative cost per kW.

VGIC respectfully offers the following recommendations to support the Board's goals of accelerating DER interconnection, increasing hosting capacity, improving affordability, and enabling EVs to operate as flexible grid resources.

- **Recommendation #1: Evaluate and plan for the opportunity to leverage the latent energy storage capacity of EVs** through utility planning processes, including load forecasting, distribution system planning, and resource adequacy analysis.
- **Recommendation #2: Optimize the use of vehicle-grid integration to increase hosting capacity and reduce distribution system investments** through managed

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[2] The Brattle Group, January 202, New York's Grid Flexibility Potential, Volume 1: Summary Report available at https://www.brattle.com/wp-content/uploads/2025/02/New-Yorks-Grid-Flexibility-Potential-Volume-I-Summary-Report.pdf.

[3] Massachusetts Department of Energy Resources, December 2025, Peak Potential: Load Management for an Affordable Net-Zero Grid available at https://www.mass.gov/doc/doer-peak-potential-report-and-policy-recommendations/download.

charging, bidirectional charging, and other operational strategies that improve grid utilization and defer infrastructure upgrades.

- **Recommendation #3: Adopt emerging best practices in bidirectional charging system interconnection** by establishing clear and standardized pathways for AC and DC bidirectional charging systems using existing technical standards and interconnection frameworks.
- **Recommendation #4: Develop mechanisms to efficiently manage installation, interconnection, and program administration for customers with multiple DERs at a single site** through streamlined review procedures, clear net metering treatment, and coordinated program participation requirements for multi-DER sites.

Electric Vehicles as Distributed Energy Resources

Vehicle-grid integration, often abbreviated as VGI, describes the coordinated management of electric vehicle charging and, where bidirectional capabilities are enabled, the controlled discharge of energy from vehicle batteries. EVs represent a rapidly growing and highly flexible form of electric load. Because vehicles are parked for the vast majority of the day, their charging behavior can often be adjusted in time without affecting mobility. Through appropriate software platforms, communications systems, and program designs, EV charging can be shifted from high-demand periods to times when the system has excess capacity, and electricity prices are lower. In addition, EVs paired with bidirectional charging systems can discharge energy stored in their batteries to meet site load or export to the grid when needed, while still meeting the driver's mobility needs and protecting the vehicle battery's health. With these capabilities, electric vehicles can serve as flexible grid resources that help balance supply and demand in the electric system and, specifically, manage constraints in the distribution system.

Vehicle-grid integration also aligns closely with emerging flexible service connection and flexible interconnection frameworks designed to increase utilization of existing grid infrastructure and reduce the need for costly system upgrades. Under these approaches, energy storage and flexible loads are interconnected subject to operational limits or dynamic controls that manage electricity imports and exports in response to distribution system constraints. These frameworks may be particularly valuable for fleet electrification applications, where large charging loads and bidirectional export capability can be actively managed to remain within grid constraints while avoiding or deferring costly upgrades. As a result, VGI should be viewed not only as a managed charging or export strategy, but also as an operational tool that can support flexible grid connections.

VGI technology has advanced quickly in recent years and is moving from early demonstrations toward broader deployment and adoption. The software platforms, communications systems, vehicles, and charging equipment needed to support these capabilities are now available in the marketplace and are increasingly being incorporated into offerings from automakers, charging equipment manufacturers, and software providers. As electric vehicle adoption grows, these technologies are creating the foundation for electric vehicles to operate as flexible grid resources.

Significant progress has already been made in deploying managed charging solutions. Several platform providers are working with utilities and automakers to scale programs that shift EV charging in response to grid conditions and electricity prices. Companies such as ChargeScape, EnergyHub, ev.energy, and WeaveGrid have developed software platforms that use vehicle telematics, charger data, and utility signals to coordinate charging across large numbers of vehicles. These platforms are already supporting managed charging programs in multiple states and demonstrate that flexible EV charging can be delivered reliably at scale.

In recent years, automakers and EV charging equipment providers have expanded the available set of bidirectional charging system offerings, detailed below in Table 1.

Table 1: EV and EVSE on New Jersey's roads that are offered with bidirectional charging features

Product	Notes
Nissan LEAF	- Model Year ("MY") 2013 or later. - Bidirectional charging enabled with dcbel Ara.[4]
Ford F-150 Lightning Electric	- Bidirectional charging enabled with Ford Charge Station Pro and Home Integration System.[5]
Chevrolet Bolt EV	- MY 2027. - Bidirectional charging enabled with GM Energy PowerShift.[6] - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB-certified inverter.

[4] dcbel, California Energy Commission. *California Connected Home Rebate.* Accessed January 13, 2026. https://redwds.dcbel.energy/

[5] Bill Crider. *Your Next Side Hustle Might Come from Your Parked Electric Vehicle.* Ford From the Road. October 28, 2025. https://www.fromtheroad.ford.com/us/en/articles/2025/introducing-ford-home-power-management

[6] GM Energy. *V2H-Capable GM EVs.* Accessed January 13, 2026. https://gmenergy.gm.com/

Chevrolet Silverado EV	- MY 2024, 2025, and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB-certified inverter.
Chevrolet Equinox EV	- MY 2024, 2025, and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB-certified inverter.
Chevrolet Blazer EV	- MY 2024, 2025, and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB-certified inverter.
GMC Hummer EV SUT	- MY 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB-certified inverter.
GMC Sierra EV	- MY 2024, 2025, and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB-certified inverter.
Cadillac CELESTIQ	- MY 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB-certified inverter.
Cadillac VISTIQ	- MY 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB-certified inverter.
Cadillac ESCALADE IQ	- MY 2025 and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB-certified inverter.

Cadillac OPTIQ	- MY 2025 and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB-certified inverter.
Cadillac LYRIQ	- MY 2024, 2025, and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB-certified inverter.
Tesla Cybertruck	- Bidirectional charging enabled with Tesla Universal Wall Connector.[7]
Kia EV9	- Bidirectional charging enabled with Wallbox Quasar 2.[8]
Volvo EX90	- Bidirectional charging enabled with dcbel Ara.[9]
Polestar 3	- Bidirectional charging enabled with dcbel Ara.[10]
BlueBird	- 155 kWh or 196 kWh school buses[11]
BYD / RIDE	- 156 kWh, 230 kWh, or 288 kWh school buses[12]

[7] Tesla. *Powershare.* Accessed January 13, 2026. https://www.tesla.com/powershare
[8] Kia EV9 x Wallbox Quasar 2. Accessed January 13, 2026. https://wallbox.com/en_us/kia-ev9-quasar-2
[9] dcbel, California Energy Commission. *California Connected Home Rebate.* Accessed January 13, 2026. https://redwds.dcbel.energy/
[10] dcbel, California Energy Commission. *California Connected Home Rebate.* Accessed January 13, 2026. https://redwds.dcbel.energy/
[11] BusinessWire. *Blue Bird Receives Record Order of 180 Electric School Buses.* January 16, 2024. https://www.businesswire.com/news/home/20240116715215/en/Blue-Bird-Receives-Record-Order-of-180-Electric-School-Buses
[12] PG&E. *Vehicle-to-Everything (V2X) pilot program: Eligible Product List for the V2X Commercial Pilot.* https://www.pge.com/en/clean-energy/electric-vehicles/getting-started-with-electric-vehicles/vehicle-to-everything-v2x-pilot-programs.html. Retrieved January 13, 2026.

IC Bus	- 210 kWh or 315 kWh school buses[13]
Green Power Motors	- 118 kWh or 193 kWh school buses[14]
Thomas Built Buses	- 226 kWh school buses[15]
Tellus Power Green	- 20, 30, 40, or 60 kW EVSE[16]
InCharge	- 22, 44, or 66 kW EVSE[17]
Heliox	- 60 kW EVSE[18]

In addition to the above-listed currently available solutions, the following EVs are associated with announced, but not yet available, bidirectional charging features: Hyundai Ioniq 9,[19] Kia EV6,[20] and Rivian R2.[21] This totals 24 models on the market today from 14 automakers, with three additional models from three more automakers already announced for near-term release. Electric Vehicle Supply Equipment ("EVSE") manufacturers that have announced but not yet

[13] IC Bus. *Electric CE Series School Bus.* https://www.icbus.com/-/media/Project/Navistar/ICBus/ICBus/electric_ce_series_brochure_spread_format.pdf. Retrieved January 7, 2025.
[14] California HVIP. *School Bus: Vehicle Type – ZESBI $ Eligible.* https://californiahvip.org/vehicle-category/school-bus/?type=655. Retrieved January 7, 2025.
[15] Thomas Built Buses, Retrieved March 3, 2026 from https://thomasbuiltbuses.com/electric-school-buses/utility-providers/
[16] PG&E. *Vehicle-to-Everything (V2X) pilot program: Eligible Product List for the V2X Commercial Pilot.* https://www.pge.com/en/clean-energy/electric-vehicles/getting-started-with-electric-vehicles/vehicle-to-everything-v2x-pilot-programs.html. Retrieved January 7, 2025.
[17] InCharge. *Chargers & Hardware.* https://inchargeus.com/chargers/. Retrieved January 7, 2025.
[18] Heliox. *Heliox 60 kW*. https://www.heliox-energy.com/us-products/60-kw-dc-charger. Retrieved January 7, 2025.
[19] Hyundai. *Hyundai Motor Group Expands EV Energy Services with Vehicle to Grid and Vehicle to Home*. November 28, 2025. https://www.hyundainews.com/releases/4634
[20] Ibid.
[21] EV.com. *$45,000 Rivian R2 Will Power Homes With New Bidirectional Charging Technology.* November 6, 2025. https://ev.com/news/45000-rivian-r2-will-power-homes-with-new-bidirectional-charging-technology

released bidirectional charging solutions include: Emporia,[22] ChargePoint,[23] Enphase,[24] SolarEdge, and Autel.

Notably, all of the above-listed bidirectional charging systems in both the light-duty and medium/heavy-duty segments is certified to UL 1741 SB. This certification ensures compliance with IEEE 1547-2018 and facilitates safe and reliable interconnection. Nearly all of the light-duty bidirectional charging system models listed above have been enabled for bidirectional operation in parallel with the grid under initiatives in either Maryland,[25] Michigan,[26] Texas,[27] California,[28] Massachusetts,[29] and/or Connecticut.[30] The medium- and heavy-duty systems have been deployed in grid-parallel bidirectional charging mode in Virginia, Maryland, North Carolina, South Carolina, Florida, Tennessee, Illinois, Iowa, Michigan, New York, Massachusetts, Maine, Vermont, Colorado, New Mexico, Washington, Oregon, Nevada, and California.[31]

In order to allow customers to operate their bidirectional charging systems in grid-parallel mode, the relevant features must be enabled, which requires the appropriate hardware and software activation after the required utility interconnection study and agreement process is completed.

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[22] Emporia. *Coming Soon: V2X Bi-Directional Charger.* Accessed January 13, 2026. https://www.emporiaenergy.com/how-the-emporia-v2x-charger-works/

[23] ChargePoint. ChargePoint and Eaton launch breakthrough ultrafast DC V2G chargers and power infrastructure to accelerate the future of EV charging. August 28, 2025. https://www.chargepoint.com/about/news/chargepoint-and-eaton-launch-breakthrough-ultrafast-dc-v2g-chargers-and-power

[24] Enphase. *IQ Bidirectional EV Charger.* Accessed January 13, 2026. https://enphase.com/ev-chargers/bidirectional

[25] Sunrun. *Sunrun and BGE Operate Nation's First Residential Vehicle-To-Grid Distributed Power Plant Using Ford F-150 Lightning Trucks,* September 24, 2025. https://investors.sunrun.com/news-events/press-releases/detail/352/sunrun-and-bge-operate-nations-first-residential

[26] Bill Crider, Ford Motor Company. *Your Next Side Hustle Might Come from Your Parked Electric Vehicle.* October 28, 2025. https://www.fromtheroad.ford.com/us/en/articles/2025/introducing-ford-home-power-management.

[27] Tesla. *What is Powershare Grid Support.* Accessed March 18, 2026. https://www.tesla.com/support/powershare/what-is-powershare-grid-support

[28] See, for example: Volvo/dcbel. *Powerful. Even when Parked.* Accessed March 18, 2026. https://volvo.dcbel.energy/; Bidirectional Energy. *Join Bidirectional Energy's V2X Charger Program in California.* Accessed March 18, 2026. https://www.bidirectional.energy/learn/california; PG&E. *Vehicle-to-Everything (V2X) pilot program.* Accessed March 18, 2026. https://www.pge.com/en/clean-energy/electric-vehicles/getting-started-with-electric-vehicles/vehicle-to-everything-v2x-pilot-programs.html

[29] Massachusetts Clean Energy Center. *Vehicle-to-Everything Demonstration Projects.* Accessed March 18, 2026. https://www.masscec.com/masscec-focus/clean-transportation/electric-vehicle-charging-infrastructure/vehicle-to-everything-demonstration

[30] Bidirectional Energy. *Join Bidirectional Energy's V2X Charger Program in Connecticut.* Accessed March 18, 2026. https://www.bidirectional.energy/learn/connecticut

[31] World Resources Institute. *Latest Lessons from Electric School Bus Vehicle-to-Grid Programs.* May 13, 2025. https://www.wri.org/update/electric-school-bus-v2g-lessons-examples

Vehicle-Grid Integration and Distribution System Planning

The recommendations in this section are responsive to the request for information's ("RFI's") focus on improving hosting capacity analysis, addressing constrained circuits, and identifying operational strategies that can support additional DER integration while minimizing infrastructure costs. Specifically, these recommendations highlight how vehicle-grid integration, including managed charging and bidirectional charging, can function as a flexible grid resource that increases effective hosting capacity, improves utilization of existing distribution infrastructure, and helps defer or reduce the need for traditional system upgrades. By incorporating VGI into distribution planning and hosting capacity analysis, the Board and utilities can better evaluate how flexible EV load and bidirectional capabilities can support the State's affordability, reliability, and grid modernization objectives.

Recommendation #1: Evaluate and Plan for the Opportunity to Leverage the Latent Energy Storage Capacity of EVs

Given the rapid advancement of vehicle-grid integration technologies, the growing availability of both managed charging platforms and bidirectional charging systems, and the increasing penetration of EVs capable of providing flexible grid services, VGIC recommends that the BPU initiate a formal process to quantify and evaluate the VGI resource in New Jersey. As demonstrated above, EVs are no longer simply an incremental load; they represent a scalable, dispatchable, and highly flexible resource capable of shifting demand and, where enabled, discharging energy to support grid reliability and manage distribution system constraints. These capabilities may be particularly valuable in emerging flexible service connection frameworks, especially for fleet electrification applications where charging and bidirectional export can be actively managed to operate within distribution system constraints and reduce the need for costly infrastructure upgrades. However, without a structured effort to quantify this resource and assess its operational characteristics, the State risks underutilizing a key asset to reduce capacity costs and place downward pressure on rates.

VGIC recommends that the BPU direct utilities, in collaboration with stakeholders, to take a practical step toward evaluating how EVs can function as grid resources, including both managed charging and bidirectional capabilities. At a minimum, utilities should be required to assess how these resources can be incorporated into core planning functions such as load forecasting, distribution system planning, and resource adequacy. This effort should include targeted analysis and stakeholder engagement, with the objective of establishing a clear foundation for integrating VGI into planning and capturing its potential to reduce costs, defer infrastructure investment, and support grid reliability.

Recommendation #2: Optimize the Use of VGI to Increase Hosting Capacity and Reduce Distribution System Investments

The electric distribution companies (EDC) responses to the RFI raised valid concerns about the operational challenges associated with high levels of distributed energy resources, particularly related to reverse power flow, voltage regulation, and system protection. For example, Jersey Central Power & Light identifies reverse power flow as a key constraint that can trigger the need for system upgrades.[32] Similarly, PSE&G notes that high penetrations of DERs can create voltage and reliability issues that may require costly infrastructure investments if not properly managed.[33] These concerns are important, but they largely reflect a static view of grid operations that assumes limited ability to actively manage load and generation, including through the use of power import limits (”PIL“) and power export limits (”PEL”).

This perspective also contributes to a broader utility concern that DERs, including vehicle-grid integration resources, are inherently expensive to integrate. For example, PSE&G anticipates that enabling large-scale DER adoption will require significant investment in traditional infrastructure and suggests that controllable resources, such as VPPs, may entail high costs.[34] While infrastructure investment will play a role, this framing does not fully reflect the growing body of evidence showing that flexible demand and bidirectional charging can reduce or defer those costs.

The body of academic and technical literature, including measurement and evaluation of utility pilots and programs, is clear that VGI, including both managed charging and bidirectional charging, can lower overall system costs when properly designed. At the system level, studies consistently find that these resources reduce peak demand, defer generation and distribution investments, improve utilization of existing infrastructure, and better align load with renewable generation. For example, planning analyses in California, New York, and Massachusetts all show that bidirectional charging can unlock gigawatts of additional capacity and billions of dollars in avoided system costs.[35] These findings directly address the concerns raised in EDC filings by demonstrating that the same constraints utilities identify can often be mitigated through operational strategies rather than capital investment.

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[32] Jersey Central Power & Light, March 5, 2026, Comments of Jersey Central Power & Light Company in response to the Notice of Request for Information In the Matter of Developing Integrated Distributed Energy Resource Plans to Modernize New Jersey’s Electric Grid BPU Docket No. QO24030199, pp. 6–10.
[33] PSE&G NJ, March 5, 2026, In the Matter of Developing Integrated Distributed Energy Resource Plans to Modernize New Jersey’s Electrical Grid Docket No. QO24030199, page 6.
[34] PSE&G, March 5, 2025, In the Matter of Developing Integrated Distributed Energy Resource Plans to Modernize New Jersey’s Electrical Grid Docket No. QO24030199, page 6.
[35] Needs multiple citations

Importantly, the literature also shows that managed charging and bidirectional charging should be viewed as complementary tools. Initial comments submitted in this docket by WeaveGrid highlight numerous studies demonstrating that VGI can create meaningful consumer value and expand hosting capacity.[36] These studies find that system benefits, particularly at the secondary distribution level, can translate into meaningful customer value, with EV owners potentially generating $100s to over $1,000 per year per vehicle in system benefits depending on program design and market conditions. This reinforces the importance of designing programs that align customer incentives with system needs.

Accessing this grid value can entail launching new programs and/or rate opportunities, which can range from simple event-based dispatch (e.g., via email notification) to real-time DER integration in a utility DERMS platform.

Given this evidence, program design and regulatory policy in New Jersey should prioritize operational solutions that actively manage load and generation in real-time. Integrating VGI into distribution planning provides a lower-cost pathway to increase hosting capacity, accommodate load growth, and support DER adoption while maintaining reliability. It also aligns with the State's broader goals of affordability and clean energy by reducing the need for costly infrastructure investments and creating new value streams for customers.

Bidirectional Charging System Interconnection and Multi-DER Integration

The recommendations in this section are responsive to the RFI's focus on energy storage system interconnection, smart inverter functionality, interoperability standards, and flexible interconnection protocols for DERs on constrained circuits. Specifically, these recommendations address the need for clear and standardized pathways to interconnect bidirectional charging systems, appropriate application of IEEE 1547 and UL 1741 standards, and coordinated treatment of hybrid DER configurations that combine solar, storage, and bidirectional EV charging. The recommendations also support the Board's interest in operational flexibility and export management by outlining practical approaches to streamline interconnection review, manage multiple DER technologies behind the meter, and ensure that program administration and tariff treatment evolve alongside increasingly integrated customer energy systems.

[36] WeaveGrid, Mrach 5, 2025, In the Matter of Developing Integrated Distributed Energy Resource Plans to Modernize New Jersey's Electric Grid, Docket No. QO24030199.

Recommendation #3: Adopt Emerging Best Practices in Bidirectional Charging System Interconnection

VGIC recommends the EDCs adopt easily implementable guidance based on observations across the U.S. to provide clear, consistent pathways to safely and reliably interconnect both types of bidirectional charging systems: Direct Current ("DC" – where the inverter is off-board the vehicle) and Alternating Current ("AC" – where the power conversion takes place onboard the vehicle). The former is most common today, with available products from Ford, General Motors, Nissan, Volvo, Polestar, Kia, dcbel, Wallbox, BlueBird, Thomas Built, RIDE, Nuvve, Heliox, InCharge and Tellus Power Green. The latter is less common but emerging, with available products from Tesla and likely other solution providers soon. VGIC respectfully urges the BPU to adopt the following general guidance for the safe and reliable interconnection of bidirectional charging systems:

a. Define Bidirectional Charging Systems as "a combination of hardware and software in or around the charger and electric vehicle for the purposes of communication with and programmed flow of energy into and out of the vehicle battery in support of utility-connected electrical loads offboard the vehicle." This definition comprehensively encompasses both DC and AC systems, without being overly prescriptive to the use case for which these systems will be enabled.
b. Clarify that the manufacturer or manufacturer-authorized party shall not enable bidirectional charging systems to operate in bidirectional mode while connected to the grid without first receiving permission to operate from the utility.
c. Clarify that bidirectional charging systems without enabled bidirectional charging features should not be directed to move through an interconnection review and study process.
d. Establish a clear pathway (i.e., an interconnection form and related guidance to installers and customers) for DC bidirectional charging systems that meet IEEE 1547-2018 via UL 1741 SB certification.
e. Establishes two pathways for AC bidirectional charging systems, in recognition that both are capable of ensuring safety and reliability, but the market has not yet coalesced around a single pathway. These two pathways for AC interconnection are suited for different use cases. Specifically, a UL 1741 SC Pathway can support multiple EVs being used at a certified charging station, whereas a UL 1741 SB CRD for DER Systems can support only a single defined EV used as part of a certified bidirectional charging system, for example, in a single-family home context.

Establishing these rules is important for expanding safe and reliable access to available VGI technology, which will lead to significant benefits for participating customers and reduce the cost impacts on the grid and, in turn, all customers

Recommendation #4: Develop Mechanisms to Efficiently Manage Multiple-DER Installations for Interconnection and Program Administration

As more customers install combinations of rooftop solar, stationary batteries, and bidirectional EV charging systems, there is a growing need to address how multiple DERs are evaluated for interconnection and how they are treated across different programs. Today, these systems are often reviewed and administered separately, which can create confusion, inconsistent requirements, and unnecessary complexity for both customers and utilities.

Utilities have already recognized this challenge. Jersey Central Power & Light notes that energy storage systems require additional review to understand charging and discharging behavior, operating modes, and controls, particularly when paired with other DERs.[37] This highlights the need for a more integrated approach to interconnection review as multi-DER configurations become more common.

Utilities and regulators have raised concerns about maintaining the integrity of net metering programs when customers add bidirectional charging systems behind the meter. These concerns are manageable. However, it is important that these issues are addressed prior to launching bidirectional EV programs, as experience in other jurisdictions suggests that approximately 50 to 70 percent of customers interested in bidirectional charging also have rooftop solar installations. A range of practical approaches exists, some of which can be implemented alongside one another, including, but not limited to:

- Treating non-exporting bidirectional EV charging additions to existing rooftop solar as non-material modifications, requiring only notification to the utility rather than a new interconnection application;
- Allowing standalone export from EV batteries under separate, non-net-metering tariffs or programs designed to encourage grid services;
- Using controls such as power control systems to ensure that site-level exports do not exceed solar generation;

[37] Jersey Central Power & Light, March 5, 2026, Comments of Jersey Central Power & Light Company in response to the Notice of Request for Information In the Matter of Developing Integrated Distributed Energy Resource Plans to Modernize New Jersey's Electric Grid BPU Docket No. QO24030199, p 8.

- Applying billing approaches that distinguish between solar and non-solar exports, including subtractive billing (e.g., total exports minus EV discharge *or* total exports minus solar generation) or submetering at the EV or charger level; and
- Using alternative methods to determine net metering credits, such as sub-metered or estimated solar generation, rather than relying solely on net meter readings.

Providing clear guidance on how these configurations will be treated will reduce uncertainty and avoid delays as customers adopt combined solar, storage, and bidirectional EV charging systems.

VGIC respectfully urges the Board to direct utilities to develop clear and standardized procedures for reviewing these configurations. This should include common combinations such as bidirectional EV charging plus solar, bidirectional EV charging plus storage, and solar plus storage plus bidirectional EV charging systems. The goal should be to create a consistent and predictable process that can be applied across utilities and projects.

Finally, the Board should ensure that program administration keeps pace with these changes. Customers with multiple DERs should be able to understand and enroll in programs, provide grid services, and receive compensation without duplicative applications, conflicting requirements, or delays. Clear rules for metering, data access, and participation will be critical to enabling these systems to deliver value to both the grid and the customer.

Conclusion

VGIC appreciates the opportunity to provide these comments and looks forward to collaboration with industry stakeholders to advance the BPU's goals of improving the efficiency and speed of DER interconnection, increasing hosting capacity on constrained circuits, and supporting the accelerated deployment of distributed energy resources to address reliability and affordability challenges in New Jersey.

Sincerely,

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